Faculty of Engineering and Technology

Department of Electrical and Computer Engineering

ENCS3340

Project 2
Comparative Study of Image Classification Using Decision Tree, Naive Bayes, and Feedforward Neural Networks

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Introduction:

This project utilizes the Fashion-MNIST dataset to examine three machine learning techniques for image classification: Decision Trees, Naive Bayes, and Feedforward Neural Networks. In our project, we employ the Fashion-MNIST dataset, which is more sophisticated than the original MNIST dataset and consists of greyscale photos of clothing items. The aim is to examine and compare the performance of these algorithms. Image classification is essential for fields like e-commerce, autonomous cars, and medical diagnosis. The results show each model's performance and which is better.

Dataset Description:

In this project, we replace the traditional MNIST dataset with a carefully selected subset of Fashion-MNIST, which contains 28×28 grayscale images of clothing items created by Zalando Research. The full dataset includes 60,000 training and 10,000 test images across 10 classes. For our work, we selected three visually distinct and practically relevant classes: T-shirt/top (class 0), Trouser (class 1), and Sandal (class 5). To ensure a balanced and computationally efficient setup, we applied stratified sampling to extract 500 training samples (approximately 167 per class) and 300 test samples (100 per class). During preprocessing, we filtered the dataset to include only the selected classes, remapped the labels to 0, 1, and 2, normalized the pixel values to the [0, 1] range, and flattened each 28×28 image into a 1D vector of length 784. This 784-dimensional vector represents each image as a single array of pixel intensities (28×28 = 784), making it suitable for input into traditional machine learning models. The resulting dataset is both balanced and standardized, supporting equal evaluation of all classification algorithms used in this study.

Table 1: Dataset Description

Metric	Training Set	Test Set
Total Samples	500	300
Samples per Class	~167	100
Image Dimensions	28×28 pixels	28×28 pixels
Feature Vector Size	784	784
Pixel Value Range	[0, 1]	[0, 1]

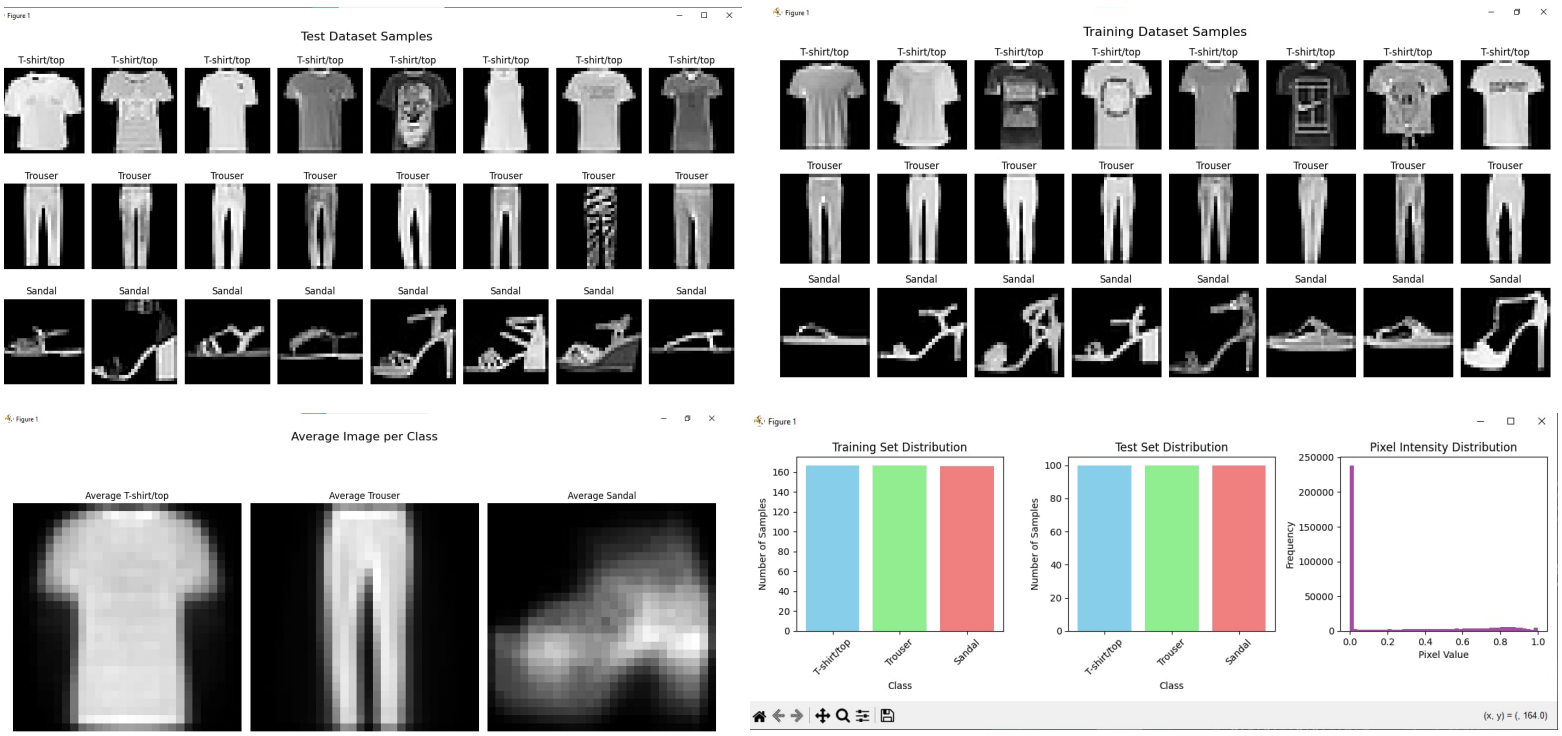

Figure 1: Result of the Dataset visualization

Model Descriptions:

- **Naive Bayes Classifier:**

  The Naive Bayes classifier, specifically GaussianNB, is based on Bayes' Theorem, which estimates the probability of a class given input features by assuming conditional independence between those features. This "naive" assumption makes computing easier by treating every pixel as independent, although nearby pixels in images are often connected. The model assumes that each feature has a Gaussian distribution per class since Fashion-MNIST pixel values are continuous and normalised to the [0,1] range. This means the likelihood of a pixel value is calculated using a normal distribution defined by the class-specific mean and variance. The addition of a tiny smoothing number avoids numerical problems caused by zero variances. Despite its simplicity, the model is very fast to train and predict, handles small datasets well, and is resilient to irrelevant features. However, because of its independence assumption, it struggles to capture complex spatial patterns in images. In Fashion-MNIST, it still achieves a reasonable accuracy since the model can take

advantage of the unique pixel patterns of some classes, such as sandals and pants, which feature horizontal straps in sandals and vertical alignment in pants.

- **Decision Tree Classifier:**

The Decision Tree classifier builds a hierarchical model by recursively splitting the feature space based on feature values that aim to maximize class separation. This is done using impurity measures such as Gini impurity, which quantifies how often a randomly chosen sample would be incorrectly labeled if randomly assigned according to the distribution of labels in a node. The Gini impurity is the splitting criterion employed in this project; hyperparameters are set to a maximum depth of 10 to avoid overfitting, and a minimum of 5 samples is needed to split a node, guaranteeing significant choices. For reproducibility, a fixed random state (42) is employed. Decision Trees are easy to interpret, require no assumptions about feature distributions, and can model complex, non-linear decision boundaries through axis-aligned splits. However, they may have issues with features that are not aligned with the axes and might overfit on small datasets. Using our Fashion-MNIST subset (T-shirt/top, Trouser, Sandal), the model learns to focus on important pixel thresholds, such as horizontal straps in sandals or vertical seams in trousers. Although it lacks spatial invariance, it performs well by leveraging the most discriminative pixel features.

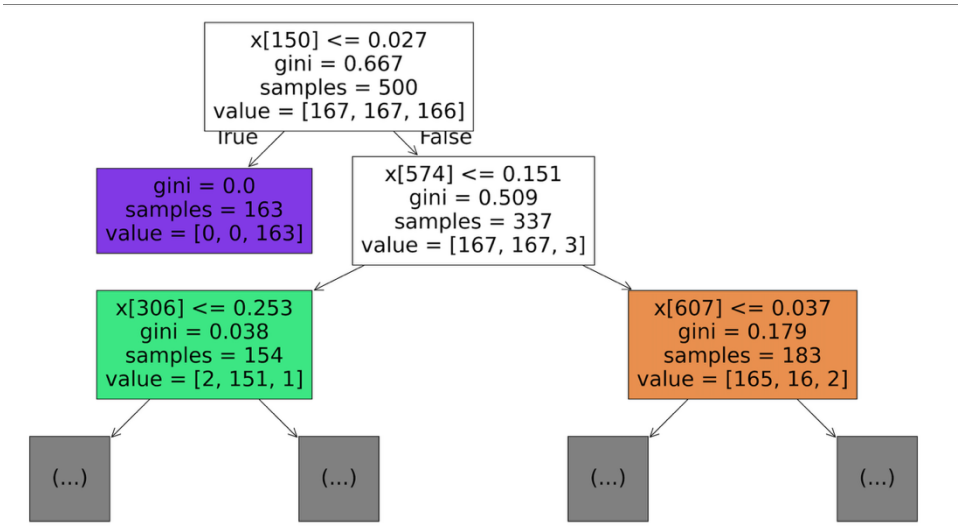

Figure 2: Decision Tree plot

- **Feedforward Neural Network:**

  The Feedforward Neural Network, also known as a Multilayer Perceptron (MLP), is designed to learn complex patterns through a sequence of layers that apply both linear and non-linear transformations. In this project, each 28×28 grayscale image is first flattened into a 784-element input vector. The model then processes this data through two hidden layers containing 128 and 64 units, respectively, each using the ReLU activation function to introduce non-linearity. The last layer generates class probabilities for the chosen apparel categories using three output nodes with softmax activation. Training is performed using the Adam optimizer, which adapts the learning rate to accelerate convergence, and the cross-entropy loss function, which penalizes incorrect predictions. Early stopping is employed to halt training when validation performance stops improving, while L2 regularization helps prevent overfitting. Although this type of model requires more data and computational resources, it excels at automatically learning multi-level features (from basic edges to full shapes) without manual intervention. The network effectively recognises differences on the Fashion-MNIST dataset, including the vertical alignment of trousers, the curved outlines of T-shirts, and the unique strap patterns of sandals. This model has the best accuracy in our analysis, although it is less interpretable than simpler algorithms.

Result:

Three machine learning methods were experimentally evaluated on a subset of the Fashion-MNIST dataset, and the results indicated disparities in performance. The Feedforward Neural Network demonstrated its capacity to recognize intricate picture patterns by achieving an accuracy of 96.67% with similarly strong precision, recall, and F1-score. The Decision Tree, which offered a balance between interpretability and performance, came in second with an accuracy of 95.67%. Naive Bayes, although simpler, still delivered solid results with 92.67% accuracy, proving effective in low-resource settings. While sandal photos were the most easily classified (99–100%

accuracy) across all models, T-shirt and pants misclassifications were the most common, especially for Naive Bayes.

```
======================================================================
Comparative Analysis
======================================================================
Model                    Accuracy   Precision  Recall     F1-score
Naive Bayes              0.9267     0.9340     0.9267     0.9262
Decision Tree            0.9600     0.9608     0.9600     0.9596
Neural Network           0.9667     0.9666     0.9667     0.9666
```

Figure 3: Comparative analysis from the code

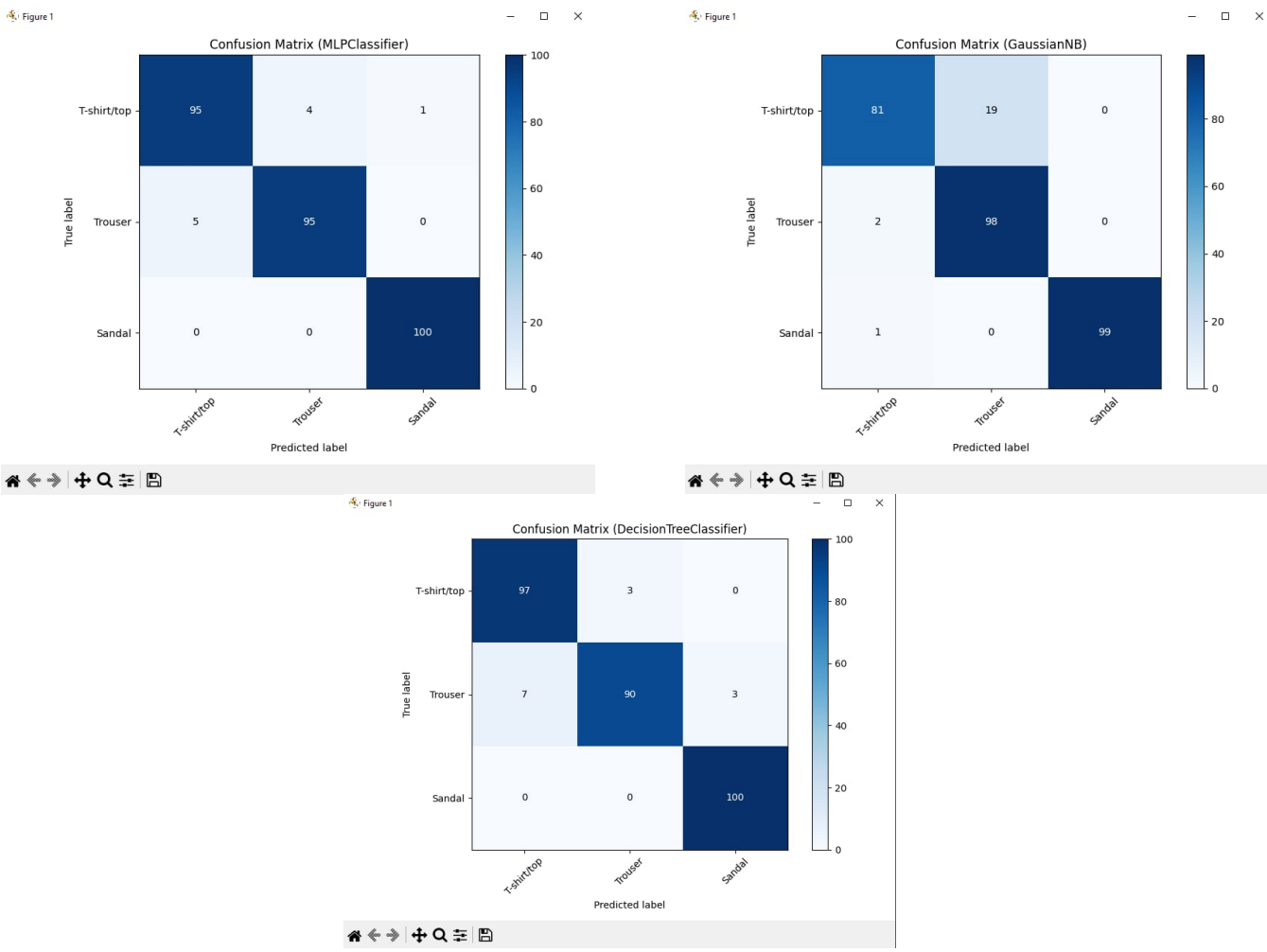

Figure 4: Confusion Matrix result

Comparative Analysis and Model Selection:

The analysis shows that the Feedforward Neural Network is the best model for classifying fashion images, with a 96.67% accuracy and strong, consistent performance in all three clothing categories. Its ability to learn hierarchical features allows it to capture delicate visual differences, especially between similar items like T-shirts and trousers. The model shows balanced misclassifications and perfect sandal classification, making it a strong candidate for deployment where high accuracy is essential. The Decision Tree, on the other hand, performed well but had poorer trouser categorisation (95.67% accuracy), suggesting problems in capturing fine-grained information. Naive Bayes (92.67% accuracy) excelled in trouser detection but struggled with T-shirts, due to its assumption of feature independence. Though less accurate, these models are still useful: Naive Bayes for its speed and low computational cost, and Decision Trees for their interpretability.

Conclusion:

This project evaluated three machine learning models (Feedforward Neural Network, Decision Tree, and Naive Bayes) for fashion image classification using a subset of the Fashion-MNIST dataset. The Neural Network achieved the highest accuracy (96.67%), effectively capturing complex visual features, while the Decision Tree (95.67%) offered interpretability, and Naive Bayes (92.67%) provided a fast, lightweight solution. All models performed exceptionally well on sandal classification (99–100%), with more variation on T-shirts and trousers. These models can be used to solve image classification problems other than Fashion-MNIST, like medical imaging, handwriting recognition, or product classification, based on the trade-offs between computational efficiency, explainability, and accuracy. This makes them valuable tools for a wide range of applications in both research and industry.